

Problem 1: trace distance

$$D(\rho, \delta) = \frac{1}{2} \text{Tr}[\rho - \delta] = \frac{1}{2} \text{Tr}[\sqrt{(\rho - \delta)^+ (\rho - \delta)}]$$

2) Solution

For any operator A

$$D(\rho, \delta) = \frac{1}{2} \text{Tr}[\rho - \delta]$$

$$\|A\| = \text{Tr}[\sqrt{A^2}]$$

$$2D(\rho, \delta) = \text{Tr}[\rho - \delta]$$

$$\text{Tr}[\sqrt{A^2}] \geq 0$$

$$\text{Tr}[\rho - \delta] \geq 0$$

$$2D(\rho, \delta) \geq 0$$

$$D(\rho, \delta) \geq \frac{0}{2}$$

$$\underline{\underline{D(\rho, \delta) \geq 0}} \quad \text{lower bounded}$$

$$\text{Tr}[\rho - \delta] = \text{Tr}[\rho + (-\delta)]$$

Triangle Inequality

$$\text{Tr}[\rho + (-\delta)] \leq \text{Tr}[\rho] + \text{Tr}[-\delta]$$

$$\leq \text{Tr}[\rho] + \text{Tr}[\delta]$$

$$\leq 1 + 1$$

$$\leq 2$$

$$\underbrace{\text{Tr}[\rho + (-\delta)]}_{= 2D(\rho, \delta)} \leq 2$$

$$2D(\rho, \delta) \leq 2$$

$$D(\rho, \delta) \leq \frac{2}{2}$$

$$D(\rho, \delta) \leq 1 //$$

$$0 \leq D(\rho, \delta) \leq 1$$

2) Solution

If $\rho = \delta$ then $D(\rho, \delta) = 0$

$$D(\rho, \delta) = \frac{1}{2} \text{Tr}[\rho - \delta]$$

$$= \frac{1}{2} \text{Tr}[\rho - \rho]$$

$$= \frac{1}{2} \text{Tr}[0]$$

$$= \frac{1}{2} \text{Tr}[0]$$

$$= \frac{1}{2} \cdot 0$$

$$= 0$$

If $D(\rho, \delta) = 0$ then $\rho = \delta$

$$D(\rho, \delta) = \frac{1}{2} \text{Tr}[\rho - \delta]$$

$$0 = \frac{1}{2} \text{Tr}[\rho - \delta]$$

$$\text{Tr}[\rho - \delta] = 0$$

$$\text{Tr}[\rho - \delta] = 0$$

$$\rho - \delta = 0$$

$$\rho - \delta = 0$$

$$\rho = \delta$$

$$\underline{\underline{\rho = \delta}}$$

$D(\rho, \delta) = 1$ if $\rho \perp \delta$

$$D(\rho, \delta) = \frac{1}{2} \text{Tr}[\rho - \delta]$$

$$= \frac{1}{2} \text{Tr}[\rho - \delta]$$

$$= \frac{1}{2} \text{Tr}[\rho + \delta]$$

$$= \frac{1}{2} [\text{Tr}[\rho] + \text{Tr}[\delta]]$$

$$= \frac{1}{2} [1 + 1]$$

$$= \frac{1}{2} [2]$$

$$= \frac{2}{2}$$

$$= 1$$

Orthogonal supported state

$$\rho - \delta = \rho + \delta$$

$$\text{Tr}[\rho] = \text{Tr}[\delta] = 1$$

3) Solution

$$D(\mathcal{S}, \delta) \leq D(\mathcal{S}, \mathcal{T}) + D(\mathcal{T}, \delta)$$

$$D(\mathcal{S}, \delta) = \frac{1}{2} \text{Tr} [|\mathcal{S} - \delta|]$$

Where add max we can put as the following

$$D(\mathcal{S}, \delta) = \frac{1}{2} \max_{\pi} \text{Tr} [\pi(\mathcal{S} - \delta)]$$

where,

$$\mathcal{S} - \delta = \mathcal{S} - \mathcal{T} + \mathcal{T} - \delta$$

$$\mathcal{S} - \delta = (\mathcal{S} - \mathcal{T}) + (\mathcal{T} - \delta)$$

$$D(\mathcal{S}, \delta) = \frac{1}{2} \max_{\pi} \text{Tr} [\pi(\mathcal{S} - \mathcal{T} + \mathcal{T} - \delta)]$$

$$= \frac{1}{2} \max_{\pi} \text{Tr} [\pi(\mathcal{S} - \mathcal{T}) + \pi(\mathcal{T} - \delta)]$$

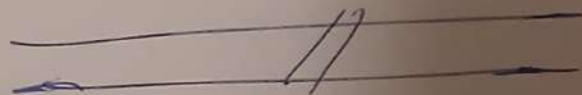
Triangle Inequality

$$\frac{1}{2} \max_{\pi} \text{Tr} [\pi(\mathcal{S} - \mathcal{T}) + \pi(\mathcal{T} - \delta)] \leq \frac{1}{2} \max_{\pi} \text{Tr} [\pi(\mathcal{S} - \mathcal{T})] + \frac{1}{2} \max_{\pi} \text{Tr} [\pi(\mathcal{T} - \delta)]$$

$$\frac{1}{2} \max_{\pi} \text{Tr} [\pi(\mathcal{S} - \mathcal{T} + \mathcal{T} - \delta)] \leq \frac{1}{2} \max_{\pi} \text{Tr} [\pi(\mathcal{S} - \mathcal{T})] + \frac{1}{2} \max_{\pi} \text{Tr} [\pi(\mathcal{T} - \delta)]$$

$$\underbrace{\frac{1}{2} \text{Tr} [|\mathcal{S} - \delta|]}_{= D(\mathcal{S}, \delta)} \leq \underbrace{\frac{1}{2} \text{Tr} [|\mathcal{S} - \mathcal{T}|]}_{= D(\mathcal{S}, \mathcal{T})} + \underbrace{\frac{1}{2} \text{Tr} [|\mathcal{T} - \delta|]}_{= D(\mathcal{T}, \delta)}$$

$$D(\mathcal{S}, \delta) \leq D(\mathcal{S}, \mathcal{T}) + D(\mathcal{T}, \delta)$$



4) Solution

$$D(\mathcal{U} \mathcal{S} \mathcal{U}^{\dagger}, \mathcal{U} \delta \mathcal{U}^{\dagger}) = \frac{1}{2} \text{Tr} [|\mathcal{U} \mathcal{S} \mathcal{U}^{\dagger} - \mathcal{U} \delta \mathcal{U}^{\dagger}|]$$

$$= \frac{1}{2} \text{Tr} [|\mathcal{U} (\mathcal{S} - \delta) \mathcal{U}^{\dagger}|]$$

$$= \frac{1}{2} \text{Tr} \left[\underbrace{U(S-\delta)U^\dagger}_{= U(S-\delta)U^\dagger} \right]$$

$$= \frac{1}{2} \text{Tr} [U(S-\delta)U^\dagger]$$

$$= \frac{1}{2} \text{Tr} [S-\delta] \underbrace{U U^\dagger}_{= I}$$

$$= \frac{1}{2} \text{Tr} [S-\delta] \cdot I$$

$$= \frac{1}{2} \text{Tr} [S-\delta]$$

$$= D(S, \delta)$$

where

$S-\delta$ is diagonal matrix

U is unitary matrix

$$U(S-\delta)U^\dagger = U(S-\delta)U^\dagger$$

$$U(S-\delta)U^\dagger = (S-\delta)U U^\dagger$$

$$\underline{\underline{U(S-\delta)U^\dagger = (S-\delta)}}$$

$$\underline{\underline{D(UBU^\dagger, U\delta U^\dagger) = D(S, \delta)}}$$

5) Given

$$a) S = \frac{3}{4} |0\rangle\langle 0| + \frac{1}{4} |1\rangle\langle 1|$$

$$\delta = \frac{2}{3} |0\rangle\langle 0| + \frac{1}{3} |1\rangle\langle 1|$$

Required

$$Q = ?$$

$$S = ?$$

$$\Pi_Q = ?$$

$$D(S, \delta) = ?$$

Solution

$$S - \delta = Q - S$$

$$S - \delta = \left(\frac{3}{4} |0\rangle\langle 0| + \frac{1}{4} |1\rangle\langle 1| \right) - \left(\frac{2}{3} |0\rangle\langle 0| + \frac{1}{3} |1\rangle\langle 1| \right)$$

$$= \left(\frac{3}{4} - \frac{2}{3} \right) |0\rangle\langle 0| + \left(\frac{1}{4} - \frac{1}{3} \right) |1\rangle\langle 1|$$

$$= \left(\frac{9-8}{12} \right) |0\rangle\langle 0| + \left(\frac{3-4}{12} \right) |1\rangle\langle 1|$$

$$= \frac{1}{12} |0\rangle\langle 0| - \frac{1}{12} |1\rangle\langle 1|$$

$$\underline{\underline{\frac{1}{12} |0\rangle\langle 0| - \frac{1}{12} |1\rangle\langle 1|}}$$

The positive part used to determine Q and S

$$Q = \frac{1}{12} 107201$$

$$S = \frac{1}{12} 117211$$

$$\Pi Q = 107201$$

$$\begin{aligned} D(3,6) &= \text{Tr}[Q] \\ &= \text{Tr}\left[\frac{1}{12} 107201\right] \\ &= \frac{1}{12} \text{Tr}[107201] \\ &= \frac{1}{12} \cdot (2) \\ &= \frac{1}{12} \end{aligned}$$

b) given

$$\begin{aligned} 3 &= \frac{3}{4} 107201 + \frac{1}{4} 117211 \\ &= \begin{pmatrix} \frac{3}{4} & 0 \\ 0 & \frac{1}{4} \end{pmatrix} \end{aligned}$$

$$\begin{aligned} 6 &= \frac{2}{3} 117211 + \frac{1}{3} 17211 \\ &= \begin{pmatrix} \frac{1}{2} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{2} \end{pmatrix} \end{aligned}$$

$$\begin{aligned} 3 - 6 &= \begin{pmatrix} \frac{3}{4} & 0 \\ 0 & \frac{1}{4} \end{pmatrix} - \begin{pmatrix} \frac{1}{2} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{2} \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{4} & -\frac{1}{6} \\ -\frac{1}{6} & -\frac{1}{4} \end{pmatrix} \end{aligned}$$

To calculate the trace distance ^{5th} calculate Q

$$Q = 1 + 14 + 7441$$

$$S = 1 - 14 - 7441$$

$$\Pi Q = 14 + 7441$$

$$\begin{aligned}
 D(8, 8) &= \text{Tr}[Q] \\
 &= \text{Tr}[(1+14+74+1)] \\
 &= 1 + \text{Tr}[14+74+1] \\
 &= 1 + 44+14+7 \\
 &= 1 + (8) \\
 &= \underline{\underline{1+}}
 \end{aligned}$$

The eigen value of Q

$$8 - 8 = \begin{pmatrix} \frac{1}{4} & -\frac{1}{6} \\ -\frac{1}{6} & -\frac{1}{4} \end{pmatrix}$$

$$\lambda_+ = +\sqrt{\left(\frac{1}{4}\right)^2 + \left(\frac{1}{6}\right)^2}$$

$$= +\sqrt{\frac{1}{16} + \frac{1}{36}}$$

$$= +\sqrt{\frac{1}{16} + \frac{1}{36}}$$

$$= +\sqrt{\frac{9+4}{144}}$$

$$= +\sqrt{\frac{13}{144}}$$

$$= +\frac{\sqrt{13}}{12}$$

$$D(9, 8) = \lambda_+$$

$$= \frac{\sqrt{13}}{12}$$

Problem 2: pure state

1) given

$$\rho = |\psi\rangle\langle\psi|$$

$$\sigma = |\varphi\rangle\langle\varphi|$$

$$|\varphi\rangle = \cos\theta|\psi\rangle + \sin\theta|\psi^\perp\rangle$$

required

$$D(\rho, \sigma) = ?$$

Solution

$$D(\rho, \sigma) = \frac{1}{2} \text{Tr}|\rho - \sigma|$$

$$\rho - \sigma = |\psi\rangle\langle\psi| - |\varphi\rangle\langle\varphi|$$

$$= |\psi\rangle\langle\psi| - (\cos\theta|\psi\rangle + \sin\theta|\psi^\perp\rangle)(\cos\theta\langle\psi| + \sin\theta\langle\psi^\perp|)$$

$$= |\psi\rangle\langle\psi| - \cos^2\theta|\psi\rangle\langle\psi| - \cos\theta\sin\theta|\psi\rangle\langle\psi^\perp| - \sin\theta\cos\theta|\psi^\perp\rangle\langle\psi| - \sin^2\theta|\psi^\perp\rangle\langle\psi^\perp|$$

$$= (1 - \cos^2\theta)|\psi\rangle\langle\psi| - \cos\theta\sin\theta|\psi\rangle\langle\psi^\perp| - \sin\theta\cos\theta|\psi^\perp\rangle\langle\psi| - \sin^2\theta|\psi^\perp\rangle\langle\psi^\perp|$$

$$= \sin^2\theta|\psi\rangle\langle\psi| - \cos\theta\sin\theta|\psi\rangle\langle\psi^\perp| - \sin\theta\cos\theta|\psi^\perp\rangle\langle\psi| - \sin^2\theta|\psi^\perp\rangle\langle\psi^\perp|$$

$$= \begin{pmatrix} \sin^2\theta & -\cos\theta\sin\theta \\ -\sin\theta\cos\theta & -\sin^2\theta \end{pmatrix}$$

$$|\rho - \sigma| = \det(\rho - \sigma)$$

$$= |\sin^2\theta(-\sin^2\theta) - (-\cos^2\theta\sin^2\theta)|$$

$$= |-\sin^4\theta - (-\cos^2\theta\sin^2\theta)|$$

$$= |-\sin^2\theta(\sin^2\theta + \cos^2\theta)|$$

$$= |-\sin^2\theta|$$

7)

$$\begin{aligned}
 |B - \emptyset| &= \det(B - \emptyset) \\
 &= |-\sin^2 \theta| \\
 &= \underline{\underline{\sin^2 \theta}}
 \end{aligned}$$

$$\begin{aligned}
 \lambda_{\pm} &= \pm \sqrt{|B - \emptyset|} \\
 &= \pm \sqrt{\sin^2 \theta} \\
 &= \pm \sin \theta \\
 &= \underline{\underline{\pm \sin \theta}}
 \end{aligned}$$

The trace norm is the sum of absolute value eigenvalue

$$\begin{aligned}
 \mathcal{D}(B, \emptyset) &= \frac{1}{2} \text{Tr} [|B - \emptyset|] = \frac{1}{2} ||B - \emptyset|| \\
 &= \frac{1}{2} ||B - \emptyset|| \\
 &= \frac{1}{2} [|\sin \theta| + |- \sin \theta|] \\
 &= \frac{1}{2} [\sin \theta + \sin \theta] \\
 &= \frac{1}{2} [2 \sin \theta] \\
 &= \sin \theta \\
 &= \underline{\underline{\sin \theta}}
 \end{aligned}$$

$$\sin \theta = \sqrt{1 - \cos^2 \theta}$$

$$\begin{aligned}
 \langle \psi | \psi \rangle &= \langle \psi | (\cos \theta |\psi\rangle + \sin \theta |\psi^\perp\rangle) = \cos \theta \underbrace{\langle \psi | \psi \rangle}_{=1} + \sin \theta \underbrace{\langle \psi | \psi^\perp \rangle}_{=0} \\
 &= \cos \theta \langle \psi | \psi \rangle + \sin \theta (0) \\
 &= \cos \theta \\
 &= \underline{\underline{\cos \theta}}
 \end{aligned}$$

$$D(\beta, \delta) = \sin \theta$$

$$= \sqrt{1 - \cos^2 \theta}$$

$$= \sqrt{1 - 1/4} = \sqrt{3}/2$$

$$D(\beta, \delta) = \sqrt{1 - 1/4} = \sqrt{3}/2$$

$$\cos 2\theta = 1/4$$

2) Given

$$D(\beta, \delta) = \sqrt{1 - 1/4} = \sqrt{3}/2$$

$$= \sqrt{1 - \cos^2 \theta}$$

$$= \sqrt{1 - \cos^2(\alpha/2)}$$

$$= \sqrt{\sin^2(\alpha/2)}$$

$$= |\sin(\alpha/2)|$$

$$= \sin(\alpha/2)$$

//

$$\theta = \frac{\alpha}{2}$$

θ = Hilbert-space angle (from overlap)
 α = Bloch-sphere angle (geometric)

The Euclidean distance $d(m, s)$ between the tips of the two Bloch vectors, r and s

$r = 1$
 $s = 1$ } for pure state

The length distance

$$d^2(m, s) = |r|^2 + |s|^2 - 2|r||s|\cos \alpha$$

$$= 1^2 + 1^2 - 2(1)(1)\cos \alpha$$

$$= 2 + 2 - 2\cos \alpha$$

$$= 2 - 2\cos \alpha$$

$$= 2(1 - \cos \alpha)$$

$$= 2(2\sin^2(\alpha/2))$$

$$= 4\sin^2(\alpha/2)$$

Half angle identity

$$1 - \cos \alpha = 2\sin^2(\alpha/2)$$

$$= 2\sin^2(\alpha/2)$$

$$d^2(r,s) = 4J \sin^2\left(\frac{\alpha}{2}\right)$$

$$d(r,s) = 2 \sin\left(\frac{\alpha}{2}\right)$$

$$\sin\left(\frac{\alpha}{2}\right) = \frac{d(r,s)}{2}$$

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$$D(3,6) = \sin\left(\frac{\alpha}{2}\right) = \frac{d(r,s)}{2}$$

$$D(3,6) = \sin\left(\frac{\alpha}{2}\right) = \frac{d(r,s)}{2}$$

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Problem 3: Dynamics

2. Given

$$z(s) = p \frac{I}{2} + (1-p)s$$

$$s = 147241$$

requires

$$D(s, z(s)) = ?$$

Solution

$$D(s, z(s)) = \frac{1}{2} \text{Tr} \| s - z(s) \|$$

$$= \frac{1}{2} \| s - z(s) \|$$

$$s - z(s) = s - \left(p \frac{I}{2} + (1-p)s \right)$$

$$= s - \left(p \frac{I}{2} + s - ps \right)$$

$$= s - p \frac{I}{2} - s + ps$$

$$= s - s - p \frac{I}{2} + ps$$

$$= ps - p \frac{I}{2}$$

$$= p \left(s - \frac{I}{2} \right)$$

//

$$\text{state} : 147 = 107$$

True state:

$$s = 147241$$

$$= 107201$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

//

$$D(s, z(s)) = \frac{1}{2} \| p \left(s - \frac{I}{2} \right) \|$$

$$= \frac{p}{2} \| \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \|$$

$$= \frac{p}{2} \| \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} \|$$

$$= \frac{p}{2} \| \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{bmatrix} \|$$

$$= \frac{p}{2} \| \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{bmatrix} \|$$

$$= \frac{p}{2} [|\frac{1}{2}| + |-\frac{1}{2}|]$$

$$= \frac{p}{2} [\frac{1}{2} + \frac{1}{2}] = \frac{p}{2} [\frac{2}{2}] = \frac{p}{2}$$

$$D(\rho, \sigma) = \frac{p}{2}$$

2) Give a physical interpretation the depolarizing channel can never exceed $\frac{1}{2}$ when the initial state is pure.

Solution

✓ The maximum possible value for p is $p=1$

✓ Completely depolarizing channel

$$E(\rho) = \frac{I}{2}$$

$$E(\rho) = p \cdot \frac{I}{2} + (1-p)\rho$$

$$= \frac{I}{2} + (0)\rho$$

$$= \frac{I}{2}$$

✓ maximum trace distance occurs when the initial pure state ρ is compared to the maximally mixed state

$$\sigma = \frac{I}{2}$$

$$D(\rho, \frac{I}{2}) = \frac{1}{2} \left\| \rho - \frac{I}{2} \right\|$$

$$= \frac{1}{2} \left\| \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \right\|$$

$$= \frac{1}{2} \left\| \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} \right\|$$

$$= \frac{1}{2} \left\| \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} \right\|$$

$$= \frac{1}{2} \left[\left| \frac{1}{2} \right| + \left| -\frac{1}{2} \right| \right]$$

$$= \frac{1}{2} \left[\frac{1}{2} + \frac{1}{2} \right] = \frac{1}{2} \left[\frac{2}{2} \right] = \underline{\underline{\frac{1}{2}}}$$

$$\underline{\underline{D_{\max} = \frac{1}{2}}}$$

The maximum value so it is never exceed $\frac{1}{2}$

$$\underline{\underline{D_{\max} = \frac{1}{2}}}$$